Switched Amplifier-Driven Nanopositioning: Integrating System Modeling and Control Tuning

Daniel Pechgraber, Johannes Wiesböck, Ernst Csencsics, *Member, IEEE*, and Georg Schitter

***Abstract*—This article presents the integrated mecha tronic and control design of a switched current amplifier-driven high-precision positioning system. An optimization-based parameter tuning process for a highly integrated state-control structure is proposed, including both dynamic requirements and target positioning uncertainty. This is enabled by a dynamic error budgeting analysis in the frequency domain, which is used to estimate position error contributions of the dominant error sources in the control system. The theoretical results are validated in practical experiments on a built prototype system, revealing a steady-state positioning uncertainty of 0.6 nm (rms) and fast reference position tracking with a rise time of 0.64 ms. Additionally, 17% less overshoot in comparison with classical loop-shaping cascade controllers is demonstrated.**

***Index Terms*—Controller tuning, dynamic error budgeting, nanometer positioning, precision, switched amplifier.**

I. Introduction

HIGH-PRECISION motion systems can be found in various applications throughout the high-tech industry, such as semiconductor manufacturing [1], atomic-force microscopy [2], and 3-D-printing [3]. Many of these systems rely on electromagnetic Lorentz-force actuation because of its favorable properties, such as the linear relation between force and current or (quasi) zero-stiffness [4], [5]. Zero-stiffness implies that vibrations originating from the stator are naturally isolated from the moving part of the actuator due to a position-independent force, which makes them well suited for high-precision applications on the nanoscale [6]. The moving part is usually constrained in the not actuated degrees of freedom by either magnetic levitation [7] or air bearings [8], to keep the zero-stiffness property, or by mechanical flexure structures [6], [9]. The latter, often the simplest and cheapest solution, can still transmit vibrations from the stator to the mover via the nonzero flexure stiffness, which is typically counteracted by a high-bandwidth position control loop [10].

The high-precision position control loops for Lorentz-force actuated positioning systems are usually implemented as cascaded-control structure incorporating an inner current control loop with a current amplifier and an outer position control loop [11]. Switched current amplifiers offer advantages over classical linear amplifiers due to their enhanced energy efficiency and minimized heat generation [11]. This is particularly important in precision positioning applications within limited-space environments, where thermal expansion effects can be a significant concern. On the other hand, they introduce additional current ripple and the need for more complex modulation schemes and current measurement procedures compared to linear current amplifiers and are therefore often avoided [12], [13].

There are many publications focusing on the development of high-precision current amplifiers alone, without taking into account their effect on the precision in a position control loop. A combination of a Luenberger estimator, a linear-quadratic regulator (LQR) for fast transient response, and an outer frequency domain controller is used for controlling a high-precision industrial current amplifier [14]. A more general approach studies the contribution of the current amplifier to the overall system precision by analyzing the error propagation through the entire mechatronic system, including the outer position control loop [15]. It takes into account offset errors, gain errors, and nonlinearity errors of the amplifier and the bandwidth of the current control loop. This investigation is further extended to current measurement noise and spurious signals occurring in the current amplifier in [16].

Especially, measurement noise coupled into the system by closing the current-control loop is considered a major cause of current amplifier-related positioning uncertainties [17]. A common approach to reduce the effects of measurement noise in a control system is the implementation of a Kalman filter (KF) as state observer, which takes two parameters, the measurement noise covariance and the process-noise covariance of

Manuscript received 24 November 2023; revised 25 March 2024 and 17 May 2024; accepted 8 July 2024. Date of publication 23 September 2024; date of current version 11 March 2025. This work was supported in part by Austrian Federal Ministry of Labour and Economy, in part by the National Foundation for Research, Technology and Development and the Christian Doppler Research Association, and in part by Micro-Epsilon Atensor GmbH and Micro-Epsilon-Messtechnik GmbH&Co. K.G.
(Corresponding author: Daniel Pechgraber.)

Daniel Pechgraber and Ernst Csencsics are with the Advanced Mechatronic Systems (AMS) Group and the Christian Doppler Laboratory for Precision Measurements in Motion, Automation and Control Institute (ACIN), TU Wien, 1040 Vienna, Austria (e-mail: pechgraber@acin.tuwien.ac.at).

Johannes Wiesböck and Georg Schitter are with the Advanced Mechatronic Systems (AMS) Group, Automation and Control Institute (ACIN), TU Wien, 1040 Vienna, Austria.
Digital Object Identifier 10.1109/TIE.2024.3429628

the system. These parameters widely depend on the problem and measurement system, and there is no commonly accepted method to tune the filter properly [18]. Mostly, it is tuned by trial-and error, or an optimization process based on real measured data is used to minimize certain performance metrics, such as the normalized estimation error squared (NEES) [19], [20]. However, the primary focus of precision positioning systems is the positioning uncertainty at the output, which is not directly used in the tuning of the controller parameters so far.

The contribution of this article is the integrated design of a switched amplifier-driven nanopositioning system and the systematic tuning of a state-control structure based on a dynamic error budget analysis in the frequency domain.

II. System Description and Modeling

The system under investigation is a one-degree of freedom (DoF) motion system actuated by a Lorentz actuator, which is driven by a switched current amplifier. In this section, the principle structure of the system is introduced and a comprehensive mathematical model is derived, considering the electrical as well as the magnetical and mechanical systems together with their dynamic coupling.

A. Switched Amplifier-Based One-DoF Positioning System

The mechanical system comprises a mover, which is suspended by a mechanical flexure structure that constrains the five nonactuated DoFs. The mover is rigidly connected to the moving part of a voice coil actuator (VCA), serving as the positioning mechanism. To enable high accelerations, the mover mass is kept as low as possible. This also showcases the achievable precision with the proposed control structure, because lighter masses are inherently more responsive to external disturbance forces, and it is therefore more challenging to minimize the position uncertainty. Fig. 1 shows the principle mechanical structure together with the circuit diagram of a switched current amplifier for driving the Lorentz actuator [14].

The amplifier has a full-bridge topology including four MOSFETs, switched corresponding to the respective duty-cycle values α_1 and α_2. This topology has the advantage that only one supply rail V_s is necessary for driving the VCA voltage in the range from $-V_s$ to $+V_s$. A small dead-time between the switching transitions ensures that no short-circuit occurs on either side. A center-aligned triangular carrier waveform is used for the PWM generation on each side as sketched in Fig. 1, leading to a symmetric on-time of each half-bridge around the center of the carrier signal. Due to this symmetry, the mean value of the actuator current i_{vca}, which is used for the control, can directly be extracted by sampling the current in the center of the triangular carrier. This eliminates the need for filtering the superimposed current ripple, which would introduce unacceptable delays in the control loop. A further advantage is that at these time instants (k, $k+1$,...) switching events in the MOSFETs only occur in the extreme duty-cycle cases $\alpha = 0$ or $\alpha = 1$, which can be avoided by design. This improves the robustness of the current measurement against electrical disturbances arising from the MOSFET switching. To reduce current ripple and EMI, an LC output filter is integrated on either half-bridge side, which makes it a viable solution for EMI-sensitive environments.

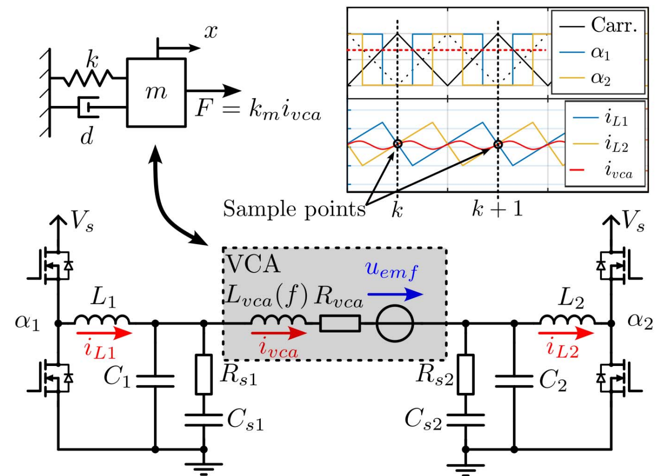

Fig. 1. Circuit diagram of switched current amplifier together with its connection to the VCA and the mechanical system represented as a mass-spring-damper system. Due to the symmetric on-times of both half-bridges around the center-aligned carrier signal, the mean value of the actuator current can be extracted without the need for additional filtering.

B. Mathematical Modeling

To derive the mathematical model for the entire electromechanical system, at first the electronic circuit from Fig. 1 has to be investigated in more detail. The VCA consists of the inductance $L_{\mathrm{vca}}(f)$, its copper resistance R_{vca} and the back-induced voltage u_{emf} originating from the movement of the VCA. Measurements show a distinct frequency dependency of the inductance, which can be explained by an eddy-current-induced magnetic flux in the stator parts of the VCA opposing the flux generated by the actuator current. The LC-output filter on both sides (L_1, C_1, L_2, C_2) reduces the current ripple in the VCA resulting from the PWM-switching. For damping the resonance peak of the LC-output filter, additional RC-snubber circuits are added (R_{s1}, C_{s1}, R_{s2}, C_{s2}).

Under the assumption of symmetric component values on both sides ($L_1 = L_2$, $C_1 = C_2$,...) and with the reduction of the two duty-cycle inputs to one generalized duty-cycle input α with $\alpha_1 = \alpha$, and $\alpha_2 = 1 - \alpha$ the full-bridge topology of Fig. 1 can be equivalently expressed by a state-space averaged (time-average over one PWM-switching period) one-sided circuit model as depicted in Fig. 2 [21]. The equivalent component values for the output filter and the snubber circuit are then calculated by $L' = 2L_1$, $C' = C_1/2$, $R'_s = R_{s1}$, and $C'_s = C_{s1}/2$. With R'_L and R'_C additionally the parasitic resistances of L' and C' are considered. The state-space averaged input voltage u_{pwm} ranges from the negative supply voltage $-V_s$ to the positive supply voltage V_s and is calculated by

$$u_{\mathrm{pwm}} = (2\alpha - 1)\, V_s \,, \alpha \in [0, 1]\,. \tag{1}$$

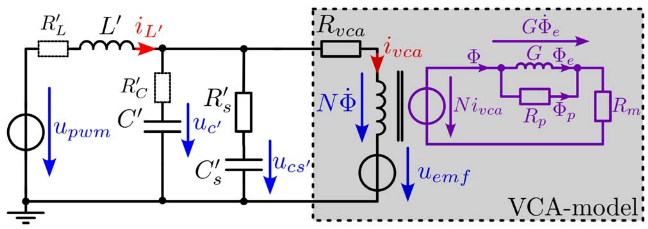

Fig. 2. Equivalent amplifier model for symmetric component values and one generalized duty-cycle input α. The frequency dependency of the VCA inductance is modeled in the magnetic domain with the main reluctance R_m, the magnetic equivalent of an inductance G representing eddy-currents, and a parallel reluctance R_p.

As shown in the grey box of Fig. 2, the VCA representation is refined to model the frequency dependency of the inductance L_{vca} originating from eddy-current effects. Therefore, the system is split into two Kirchhoffian domains: one magnetic and one electric [22]. In the magnetic domain, eddy-currents are represented by a "magnetic inductance" G, which counteracts the magnetic flux generated by the current i_{vca} together with a parasitic parallel reluctance R_p. At zero frequency, the "magnetic inductance" G has no effect, and the actuator inductance is determined with the coil windings N and the main reluctance R_m by the well-known formula $L_{\text{vca}}(0) = (N^2/R_m)$. Φ_e represents the magnetic flux through the eddy-current inductance and Φ_p the flux through the parasitic reluctance R_p. With $\Phi = \Phi_e + \Phi_p$, the coupling between the electrical and magnetical domain is given by

$$Ni_{\text{vca}} = G\dot{\Phi}_e + R_m\Phi \tag{2a}$$

$$u_{\text{vca}} = R_{\text{vca}}i_{\text{vca}} + N\dot{\Phi} + u_{\text{emf}} \tag{2b}$$

$$\Phi_p = \frac{G\dot{\Phi}_e}{R_p}\,. \tag{2c}$$

It is to note that the magnetical parameters (N, G, R_m, R_p) cannot be directly obtained from Fig. 1, so they are determined later in the system identification process (Section V-A). Further, the actuator resistance R_{vca} is subject to heating in the copper windings, which leads to a change in resistance [23]. However, as thermal effects are slow compared to the target control bandwidth, the inclusion of an integrator will cope with these changes instead of explicitly modeling it.

The mechanical subsystem of the positioning system is modeled as mass-spring-damper system with the mover mass m, spring constant k, and damping coefficient d (see Fig. 1). It is dynamically coupled to the electrical subsystem by the actuator force F and the counter-induced voltage u_{emf}. The coupling equations between magnetical and mechanical subsystems are as follows:

$$F = k_m i_{\text{vca}} \tag{3a}$$

$$u_{\text{emf}} = k_m\dot{x} \tag{3b}$$

with the actuator motor constant k_m and mover velocity $\dot{x}$. For simplicity, it is assumed that the actuator motor constant k_m is independent from the mover position x, which is valid for small motion ranges [4]. To control the position, measurements of both the actuator current i_{vca} and the mover position x are accessible.

With the states $\mathbf{x} = \begin{bmatrix}\dot{x} & x & i_{L'} & u_{c'} & u_{cs'} & \dot{\Phi}_e & \Phi_e\end{bmatrix}^T$, the input $u = u_{\text{pwm}}$, the output $\mathbf{y} = \begin{bmatrix}x & i_{\text{vca}}\end{bmatrix}^T$, $\mathbf{A} \in \mathbb{R}^{7\times7}$, $\mathbf{b} \in \mathbb{R}^{7\times1}$, and $\mathbf{C} \in \mathbb{R}^{2\times7}$, the continuous dynamic model of the entire mechatronic system can be derived in the state-space form

$$\dot{\mathbf{x}} = \underbrace{\begin{bmatrix}\mathbf{A}_{11} & \mathbf{0} & \mathbf{A}_{13}\\ \mathbf{0} & \mathbf{A}_{22} & \mathbf{A}_{23}\\ \mathbf{A}_{31} & \mathbf{A}_{32} & \mathbf{A}_{33}\end{bmatrix}}_{\mathbf{A}}\mathbf{x} + \underbrace{\begin{bmatrix}0 & 0 & \frac{1}{L'} & 0 & 0 & 0 & 0\end{bmatrix}^T}_{\mathbf{b}^T} u \tag{4a}$$

$$\mathbf{y} = \underbrace{\begin{bmatrix}0 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & \frac{G(R_m+R_p)}{NR_p} & \frac{R_m}{N}\end{bmatrix}}_{\mathbf{C}}\mathbf{x}\,. \tag{4b}$$

The dynamic matrix $\mathbf{A}$ is partitioned into the mechanical part

$$\mathbf{A}_{11} = \begin{bmatrix}-\frac{d}{m} & -\frac{k}{m}\\ 1 & 0\end{bmatrix} \tag{5}$$

the electrical part

$$\mathbf{A}_{22} = \begin{bmatrix}\frac{(-R'_s-R'_L)R'_C-R'_sR'_L}{L'(R'_C+R'_s)} & \frac{-R'_s}{(R'_C+R'_s)L'} & \frac{-R'_C}{(R'_C+R'_s)L'}\\ \frac{R'_s}{C'(R'_C+R'_s)} & \frac{-1}{C'(R'_C+R'_s)} & \frac{1}{C'(R'_C+R'_s)}\\ \frac{R'_C}{C'_s(R'_C+R'_s)} & \frac{1}{C'_s(R'_C+R'_s)} & \frac{-1}{C'_s(R'_C+R'_s)}\end{bmatrix} \tag{6}$$

and the magnetical part

$$\mathbf{A}_{33} = \begin{bmatrix}a_1 & a_2\\ 1 & 0\end{bmatrix} \tag{7}$$

with

$$a_1 = \frac{-\left((R_{vca}+R'_s)R'_C + R_{vca}R'_s\right)(R_m+R_p)}{N^2(R'_C+R'_s)} - \frac{R_p}{G} \tag{8a}$$

$$a_2 = -\frac{R_m\left((R_{vca}+R'_s)R'_C + R_{vca}R'_s\right)R_p}{G\,N^2(R'_C+R'_s)}\,. \tag{8b}$$

The dynamic coupling between the three domains results from (2) and (3) and is given by the off-diagonal matrices

$$\mathbf{A}_{13} = \begin{bmatrix}\frac{Gk_m(R_m+R_p)}{R_pNm} & \frac{R_mk_m}{Nm}\\ 0 & 0\end{bmatrix},\ \mathbf{A}_{31} = \begin{bmatrix}\frac{-R_pk_m}{NG} & 0\\ 0 & 0\end{bmatrix} \tag{9a}$$

$$\mathbf{A}_{23} = \begin{bmatrix}\frac{GR'_sR'_C(R_m+R_p)}{NR_p(R'_C+R'_s)L'} & \frac{R'_CR'_sR_m}{N(R'_C+R'_s)L'}\\ \frac{-GR'_s(R_m+R_p)}{C'NR_p(R'_C+R'_s)} & \frac{-R'_sR_m}{C'N(R'_C+R'_s)}\\ \frac{-GR'_C(R_m+R_p)}{NR_p(R'_C+R'_s)C'_s} & \frac{-R'_CR_m}{C'_sN(R'_C+R'_s)}\end{bmatrix} \tag{9b}$$

$$\mathbf{A}_{32} = \begin{bmatrix}\frac{R'_sR'_CR_p}{NG(R'_C+R'_s)} & \frac{R'_sR_p}{NG(R'_C+R'_s)} & \frac{R'_CR_p}{NG(R'_C+R'_s)}\\ 0 & 0 & 0\end{bmatrix}. \tag{9c}$$

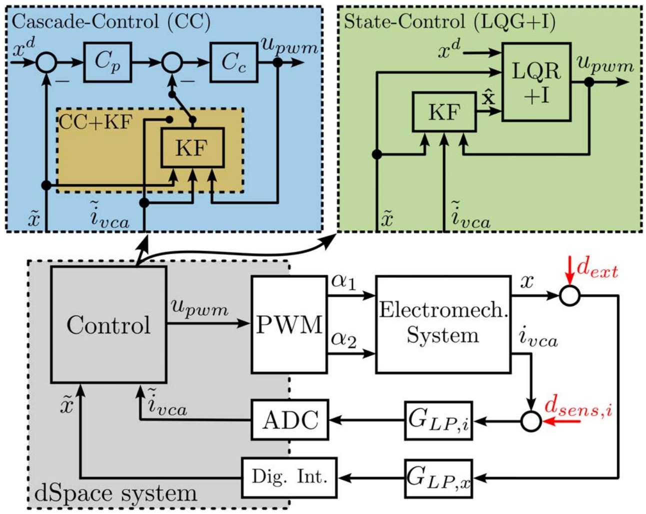

Fig. 3. Block diagram of proposed integrated-control structure and the cascade-control (CC) structure. The integrated-control structure uses an LQR with integrator (LQR+I) state-space controller in combination with a KF for the estimation of the unmeasured states (LQG+I). The cascade controller consists of the inner current controller C_c and the position controller C_p. The current measurement can either be used directly CC or in combination with the KF (CC+KF).

III. Control Design

A. Integrated State-Control Structure

Based on the system model (4), an optimal LQR state-space controller is now derived. Fig. 3 shows the proposed control structure, including the physical electromechanical system with its measured outputs x and i_{vca} and two first-order low-pass antialiasing filters ($G_{\text{LP},i}$, $G_{\text{LP},x}$). The LQR requires full-state feedback, but only two of the seven system states are directly measured. For this reason, a KF is included as a state observer.

The goal of the control is to track a reference position x^d; therefore, the LQR is formulated in the translated coordinates $\mathbf{z} = \mathbf{x} - \mathbf{x}_S$ and $u* = u - u_S$, with $\mathbf{x}_S$ and u_S being the steady-state state-vector and control input, respectively. Setting $\mathbf{0} = \mathbf{A}\mathbf{x} + \mathbf{b}u$ with (4) and solving for each state and u depending on the desired position x^d yields the following:

$$\mathbf{x}_S = \underbrace{\begin{bmatrix} 0 & 1 & \frac{k}{k_m} & \frac{R_{\text{vca}}k}{k_m} & \frac{R_{\text{vca}}k}{k_m} & 0 & \frac{Nk}{R_m k_m} \end{bmatrix}^T}_{\mathbf{X}} x^d \tag{10a}$$

$$u_S = \underbrace{k\frac{R_L + R_{\text{vca}}}{k_m}}_{U} x^d. \tag{10b}$$

To cope with model uncertainties and slow changes in system parameters, such as changing actuator resistance due to heating, an integrator is included in the LQR control design (LQR+I). The system model (4) is discretized with the sample time T_s and an integrator state z^I is included

$$\mathbf{z}^a = \begin{bmatrix} \mathbf{z} \\ z^I \end{bmatrix}, \; z^I_{k+1} = z^I_k + T_s \left(x^d_k - \tilde{x}_k \right) \tag{11}$$

forming the augmented discrete time system

$$\mathbf{z}^a_{k+1} = \mathbf{\Phi}^a \mathbf{z}^a_k + \mathbf{\Gamma}^a u_k \tag{12a}$$

$$y_k = \mathbf{C}^a \mathbf{z}^a_k. \tag{12b}$$

Taking (11) into account, the augmented state-space matrices result to

$$\mathbf{\Phi}^a = \begin{bmatrix} \mathbf{\Phi} & \mathbf{0} \\ -\mathbf{C}_1 & 0 \end{bmatrix}, \; \mathbf{\Gamma}^a = \begin{bmatrix} \mathbf{\Gamma} \\ 0 \end{bmatrix} \tag{13}$$

with $\mathbf{\Phi}$ and $\mathbf{\Gamma}$ denoting the discretized dynamic matrix and input vector of (4) and $\mathbf{C}_1$ being the first row of the matrix $\mathbf{C}$. This is, because for the integral part, only the measured position $\tilde{x}$ is considered. With a positive semidefinite matrix $\mathbf{Q} \in \mathbb{R}^{8\times 8}$ and the positive constant $R \in \mathbb{R}$, the LQR feedback gain $\mathbf{K}$ is calculated with the Matlab command *lqrd* by minimizing the cost function [24]

$$\arg\min_{u_k} J(u_k) = \sum_{k=1}^{\infty} \left(\mathbf{z}^T \mathbf{Q} \mathbf{z} + u_k^T \mathbf{R} u_k \right). \tag{14}$$

In the same manner, the KF is designed with the command *kalmd* for the discrete system model without additional integrator state. It is parameterized with the (continuous) process-noise covariance matrix $\mathbf{Q}_n \in \mathbb{R}^{7\times 7}$ and the measurement noise covariance matrix $\mathbf{R}_n \in \mathbb{R}^{2\times 2}$ delivering the estimated system states $\hat{\mathbf{x}}$ by

$$\hat{\mathbf{x}}_{\text{k}+1} = \mathbf{\Phi}\hat{\mathbf{x}}_k + \mathbf{\Gamma} u_k + \mathbf{L}\left(y_k - \mathbf{C}\hat{\mathbf{x}}_k \right) \tag{15}$$

with the KF gain matrix $\mathbf{L} \in \mathbb{R}^{7\times 2}$. Given the state estimate from the KF and by using (10), the final control output of the state-control scheme is calculated via

$$u_k = \underbrace{\begin{bmatrix} \mathbf{K}_z & K_I \end{bmatrix}}_{\mathbf{K}} \begin{bmatrix} \hat{\mathbf{x}}_k - \mathbf{X} x^d_k \\ z^I_k \end{bmatrix} + U x^d_k. \tag{16}$$

The closed-loop stability of the linear plant controlled by an LQR is given by design and the stability of the combination of LQR and KF is given by the separation theorem for linear systems [24], [25].

B. CC Structure

As a reference control structure, a conventional CC scheme is derived with a loop-shaping approach [5] in the frequency domain. As shown in Fig. 3, it consists of the outer position controller C_p and the inner current controller C_c. The current controller is comprised of a PI-controller C_{PI} and notch-filter C_n with the structure

$$C_{PI}(s) = \frac{k_{I,c}}{s}\left(1 + sT_I\right) \tag{17a}$$

$$C_n(s) = \frac{s^2 + 2D\xi\omega_n s + \omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \tag{17b}$$

according to $C_c(s) = C_{PI}(s)C_n(s)$ in the Laplace domain. The notch-filter is tuned for suppressing the resonance peak of the LC-output filter of the switched amplifier (Fig. 2) and therefore enables a higher bandwidth of the current control loop.

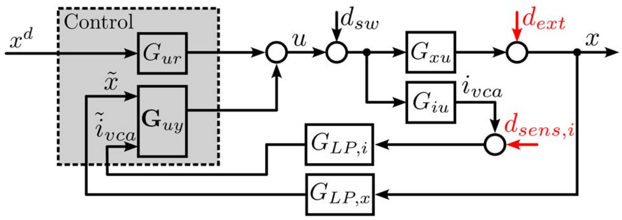

Fig. 4. System block diagram used for dynamic error budgeting with frequency domain representation of the control. The modeled error sources affect the system at the locations represented with arrows.

The position controller is implemented as PID controller with the structure [26]

$$C_p(s) = k_p + \frac{k_{I,p}}{s} + \frac{k_d s}{1 + sT_t}. \tag{18}$$

In addition to the direct measurement of the position and the current, the possibility of using the estimated current signal from the KF is added (yellow CC+KF block in Fig. 3).

IV. Position Uncertainty Analysis

To analyze the impact of disturbances and noise sources on the position uncertainty of the system, dynamic error budgeting can be used [27], [28], [29]. Every error source in the system is modeled as an (equivalent) noise source with a certain power spectral density (PSD), and its contribution to the position uncertainty is calculated with the sensitivity function $|S(j\omega)|^2$ from error location to the output x [30]. Therefore, the frequency domain representation of the controller and the plant is required for the calculation.

A. Frequency Domain Representation

As shown in Fig. 3, the derivation of the sensitivity functions in the case of CC is straightforward, as the controllers are directly available as frequency domain transfer functions. Further, the transfer functions from the voltage u to the actuator current i_{vca} and the position x can be derived from the introduced state-space model (4) via

$$\begin{bmatrix} G_{xu}(s) \\ G_{iu}(s) \end{bmatrix} = \mathbf{C}\left(s\mathbf{I} - \mathbf{A}\right)^{-1}\mathbf{b}. \tag{19}$$

In the proposed state-control case, however, the general block diagram in Fig. 3 does not intuitively lead to a frequency domain representation of the controller. It is thus redrawn as shown in Fig. 4. All control structures can be represented in the frequency domain by the transfer function G_{ur} from the reference position x^d to the control output u, and the transfer matrix $\mathbf{G}_{uy} \in \mathbb{R}^{1\times 2}$ from the measurement to the control output. In the case of LQG+I, they are derived using (10), (11), (15), and (16) and the superposition principle for linear systems. By z-transformation of (11) and (15), z^I and $\hat{\mathbf{x}}$ can be expressed by

$$z^I = \frac{1}{z-1}\left(x^d - \begin{bmatrix}1 & 0\end{bmatrix}\mathbf{y}\right) \tag{20a}$$

$$\hat{\mathbf{x}} = \mathbf{\Lambda}\mathbf{L}\mathbf{y} + \mathbf{\Lambda}\mathbf{\Gamma}u \tag{20b}$$

with $\mathbf{\Lambda} = (z\mathbf{I} - \mathbf{\Phi} + \mathbf{LC})^{-1}$. Substituting (20a) into the z-transform of (16) yields the following:

$$u = \mathbf{K}_z\left(\hat{\mathbf{x}} - \mathbf{X}x^d\right) + Ux^d + \frac{K_I}{z-1}\left(x^d - \begin{bmatrix}1 & 0\end{bmatrix}\mathbf{y}\right). \tag{21}$$

After including (20b) into (21) and solving for u, the frequency domain transfer function of the state-control structure (LQG+I) can be expressed by

$$u = \underbrace{(1-\mathbf{K}_z\mathbf{\Lambda\Gamma})^{-1}\left(\mathbf{K}_z\mathbf{\Lambda L} - \frac{K_I}{z-1}\begin{bmatrix}1 & 0\end{bmatrix}\right)}_{\mathbf{G}_{uy}=\begin{bmatrix}G_{ux}\\ G_{ui}\end{bmatrix}}\underbrace{\begin{bmatrix}\tilde{x}\\ \tilde{i}_{\text{vca}}\end{bmatrix}}_{\mathbf{y}} \tag{22}$$
$$+\underbrace{(1-\mathbf{K}_z\mathbf{\Lambda\Gamma})^{-1}\left(U + \frac{K_I}{z-1} - \mathbf{K}_z\mathbf{X}\right)}_{G_{ur}}x^d.$$

In the CC case and in the case where the CC is extended with the KF (CC+KF), the control system can also be expressed by G_{ur} and $\mathbf{G}_{uy}$. The derivation is done in a similar manner as in the demonstrated case of state-control.

B. Position Error Sources

Several error sources affect the position accuracy of the system under investigation. They include external noise sources such as power-supply noise and current measurement noise, nonlinear effects of some system components, such as PWM-switching and quantization effects, or the discrete implementation of the controllers. For reasons of simplicity, only the most prominent position error sources are considered. In Figs. 3 and 4, the location of the error sources in the control loop is shown.

The current sensor noise d_{noise} consists of the electrical noise introduced by the sensing shunt resistor and its corresponding current-sense amplifier. Based on measurements (data not shown), the current sensor noise can be approximated by a constant PSD $n_{\text{sens},i}$ (A^2/Hz). Considering that the current is sampled with the switching frequency of the current amplifier T_s, this noise value has to be multiplied with some factor k_{alias}, to account for aliased noise components with frequencies greater than the Nyquist frequency $f_N = 1/2T_s$. The cutoff frequency f_c of the first-order antialiasing filter ($G_{\text{LP},i}$ in Fig. 4) is chosen so as not to introduce significant delays in the feedback loop. For determining the factor k_{alias}, the concept of the equivalent noise bandwidth f_{enb} is used, in which the frequency behavior of $G_{\text{LP},i}$ is approximated by an ideal low pass with unity gain up to the equivalent noise bandwidth f_{enb}. For a first-order low pass it can be shown that for equivalent noise energy on the output this frequency is $f_{\text{enb}} = (\pi/2)f_c$ [31], which leads to

$$k_{\text{alias}} = \frac{f_{\text{enb}}}{f_N} = \frac{\pi}{2}f_c 2T_s = \pi f_c T_s. \tag{23}$$

The current sensor noise used for dynamic error budgeting $d_{\text{sens},i}$ follows accordingly with $d_{\text{sens},i} = k_{\text{alias}}n_{\text{sens},i}$.

The external disturbance d_{ext} is dominated by vibrations of the environment which are transmitted to the mover via the flexure stiffness. It is determined by recording the position signal

x with disabled controllers for several seconds and performing a fast Fourier transform (FFT) on the time-domain data. The worst case over multiple measurements is taken for calculating the PSD.

It is important to point out that the PWM-switching results in voltage ripple at the output of the current amplifier d_{sw}, which exhibits its first harmonic at twice the switching frequency of the PWM full-bridge (see Fig. 1). Considering the incorporation of the LC-output filter and that the suspended mover acts as a second-order mechanical low pass, the impact on the position uncertainty is negligible. This point will be further reinforced in the next section through the analysis of the error sensitivity function.

C. Error Sensitivity Functions

The error sensitivity functions represent the sensitivity of the position x to the respective error source ($d_{\text{sens},i}$, d_{ext}, d_{sw}). The derivation procedure is equivalent for all cases and is demonstrated for the current measurement noise $d_{\text{sens},i}$.

1) Set x^d, d_{sw}, and d_{ext} to zero (superposition principle).
2) Calculate the plant transfer functions $G_{xu}(s)$ and $G_{iu}(s)$ (19) and the controller transfer matrix $\mathbf{G}_{uy}$ (22).
3) Use the block diagram (Fig. 4) and solve for $x(s)$ depending on $d_{\text{sens},i}(s)$ in the frequency domain.
4) The error sensitivity function is then given by $S_{\text{sens},i}(s) = (x(s)/d_{\text{sens},i}(s))$.

The resulting error sensitivity function is given by

$$S_{\text{sens},i}(s) = \frac{G_{ui}G_{xu}G_{\text{LP},i}}{1 - G_{ui}G_{iu}G_{\text{LP},i} - G_{ux}G_{xu}G_{\text{LP},x}}. \tag{24}$$

In Fig. 5, the error sensitivity functions are plotted together with the PSD of the investigated error sources $d_{\text{sens},i}$ and d_{ext}.

The values for the system parameters and error sources are taken from the actual experimental prototype system, which is developed in Section V-A, and the controller parameter tuning is discussed in the next section. From the error sensitivity functions it is apparent that the incorporation of the KF offers a significantly enhanced capability to attenuate the prevailing current measurement noise compared to the CC case without KF (colored, solid lines). By adjusting the respective entry of the $\mathbf{R}_n$-matrix of the KF, it is possible to tune the current measurement noise sensitivity $S_{\text{sens},i}$, thus allowing for adaption of the desired noise rejection performance.

Further, it is evident that the rejection of external disturbances is similar across all three investigated cases. This similarity arises from the tuning of the control schemes to the same closed-loop bandwidth and the inclusion of an integrator in each case. Interestingly, in the CC+KF case, the external disturbance rejection is even enhanced in the lower frequency range. Conversely, the LQG+I configuration shows superior disturbance rejection within the frequency range from 200 Hz to 2 kHz, as illustrated in the zoomed section of Fig. 5.

Additionally, the sensitivity to voltage ripple from PWM-switching is plotted (colored, dashed-dotted lines), which shows a -80 dB per decade slope after the cutoff frequency of the LC-output filter. This confirms that the position uncertainty contribution of the high-frequent voltage ripple (100 kHz for the prototype setup) is negligible.

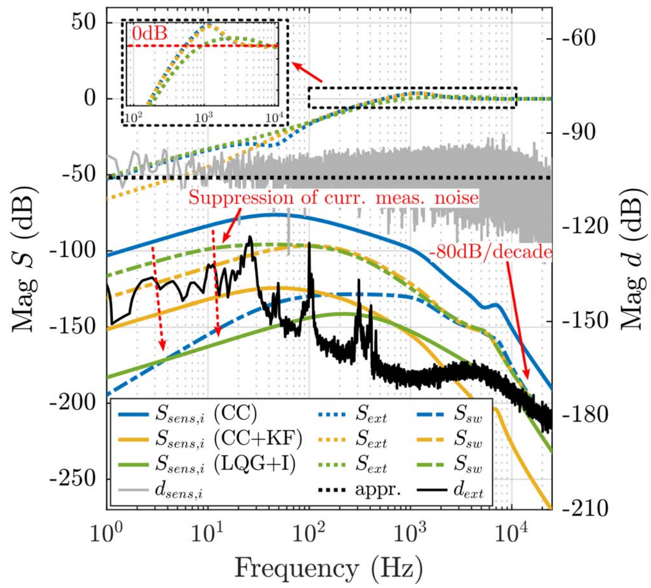

Fig. 5. Error sensitivity functions (colored, left y-axis) and PSD of current measurement noise $d_{\text{sens},i}$ (A^2/Hz) and external disturbances (m^2/Hz) (gray/black, right y-axis) for the CC, CC+KF, and LQG+I case. The suppression of current measurement noise with the KF is highlighted by the dotted red arrows.

D. Controller Parameter Tuning

Based on the system model and presented error analysis in the frequency domain, the LQR+I controller and the KF are systematically tuned to a desired performance. Therefore, the dynamic position error x_e is evaluated by applying the sensitivity functions to the PSD of the error sources. The resulting output position error PSD originating from each error source can be calculated by [30]

$$|PSD_{x,\text{sens},i}(f)| = |S_{\text{sens},i}(s)|^2 |d_{\text{sens},i}(f)| \tag{25a}$$
$$|PSD_{x,ext}(f)| = |S_{\text{ext}}(s)|^2 |d_{\text{ext}}(f)|. \tag{25b}$$

For uncorrelated error sources, the total PSD of the position error is then given by [30]

$$|PSD_{x,\text{tot}}(f)| = |PSD_{x,\text{sens},i}(f)| + |PSD_{x,\text{ext}}(f)|. \tag{26}$$

By now integrating over the relevant frequency range $[f_1, f_2]$, the rms-value of the position uncertainty in the time domain is calculated by

$$x_{e,\text{rms}} = \sqrt{\int_{f_1}^{f_2} |PSD_{x,\text{tot}}(f)|\, df}. \tag{27}$$

With this formulation of the positioning uncertainty, together with the frequency domain representation of the controller and the plant, the control objectives can be expressed in a cost function depending on the controller parameters. This allows

TABLE I
COST FUNCTION TERMS AND NORMALIZATION WEIGHTS

	Cost Function Terms	Unit	Normalization Weights	
e_1	$f_{3\text{dB}} - f_{3\text{dB}}^d$	Hz	w_1	10
e_2	$\max\left(0, x_{e,\text{rms}} - x_{e,\text{rms}}^{\max}\right)$	m	w_2	1×10^{-9}
e_3	$\lVert T(s) \rVert_\infty$	dB	w_3	1
e_4	$\lVert S(s) \rVert_\infty$	dB	w_4	2
e_5	$\lvert S(j2\pi 1\,\text{Hz}) \rvert$	1	w_5	5×10^{-3}

the formulation of the control parameter tuning process as a multidimensional optimization problem

$$\min_{\mathbf{Q}_n, \mathbf{R}_n, \mathbf{Q}, R} \quad J(\mathbf{Q}_n, \mathbf{R}_n, \mathbf{Q}, R) = \sum_{k=1}^{5} \left(\frac{e_i}{w_i}\right)^2 \tag{28}$$

with several cost terms e_i and normalization weights w_i, which are summarized in Table I. In addition to (27), the calculation of the complementary sensitivity function $T(s) = (x(s)/x^d(s))$ and the sensitivity function $S(s) = (x(s)/d_{\text{ext}}(s))$ from Fig. 4 is required for the evaluation of the cost function. It is to be noted that the dependency of the cost terms on the tuning parameters is not explicitly stated in the table for easier notation.

The principle goal of the control is to achieve a certain closed-loop bandwidth $f_{3\text{dB}}^d$ in combination with a desired maximum positioning uncertainty $x_{e,\text{rms}}^{\max}$. This is reflected in the cost function by penalizing the difference of $f_{3\text{dB}}$ from the complementary sensitivity function $T(s)$ to $f_{3\text{dB}}^d$ and values $x_{e,\text{rms}}$ above $x_{e,\text{rms}}^{\max}$. Second, to have good transient behavior and to maximize robustness, the H_∞-norms of $T(s)$ and $S(s)$, which correspond to the respective peak values in the magnitude functions, are penalized. Additionally, to ensure sufficient rejection of low-frequent disturbances and model uncertainties, the absolute value of the sensitivity function $S(s)$ evaluated at a frequency of 1 Hz is used. This term ensures that the introduced integrator in the control is tuned accordingly.

The cost function (28) is now minimized by varying the LQR+I and KF controller parameters $\mathbf{Q}_n$, $\mathbf{R}_n$, $\mathbf{Q}$, and R. Since the number of tunable parameters is 118 in this case (refer to Section III-A), the matrices are chosen as diagonal matrices

$$\begin{aligned} \mathbf{Q}_n &= \mathbf{diag}\left(\mathbf{q}_n\right), \quad && \mathbf{q}_n \in \mathbb{R}^7 \\ \mathbf{R}_n &= \mathbf{diag}\left(\mathbf{r}_n\right), && \mathbf{r}_n \in \mathbb{R}^2 \\ \mathbf{Q} &= \mathbf{diag}\left(\mathbf{q}\right), && \mathbf{q} \in \mathbb{R}^8 \\ & && R \in \mathbb{R} \end{aligned} \tag{29}$$

which reduces the optimization problem to

$$\min_{\mathbf{q}_n, \mathbf{r}_n, \mathbf{q}, R} \quad J(\mathbf{Q}_n, \mathbf{R}_n, \mathbf{Q}, R) = \sum_{k=1}^{5} \left(\frac{e_i}{w_i}\right)^2 \tag{30}$$

with 18 tunable parameters. The optimization problem is solved with an interior-point algorithm of the *fmincon* command in Matlab. The starting parameter set for the algorithm is manually chosen, considering known system properties. For example, the entries for the measurement noise covariance matrix $\mathbf{R}_n$ are chosen based on the reliability of the respective measurement.

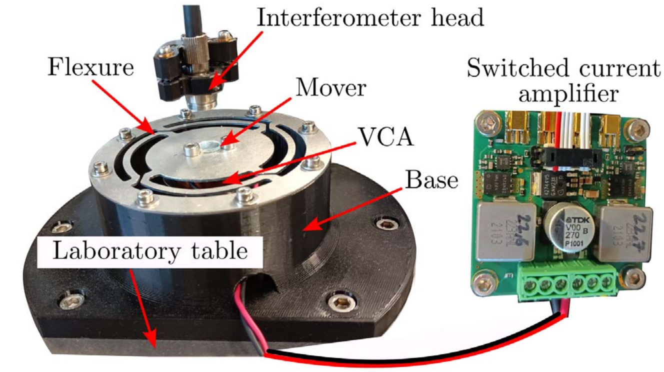

Fig. 6. High-precision one-DoF positioning system with Lorentz-type positioning mechanism, optical interferometer position measurement, and custom-made switched current amplifier.

Further, the entries in $\mathbf{Q}$ corresponding to the position x and the integrated position error [states 2 and 8 of (12)] are weighted higher than the other entries.

V. EXPERIMENTAL VALIDATION

To validate the derived system model and to evaluate the control structures with the optimized control parameters, an experimental prototype system of a one-DoF precision positioning system is developed according to Section II-A. For this system, the theoretical dynamic error budget is evaluated with the derived sensitivity functions and compared to measurement data. Afterward, the dynamic behavior of the integrated-control structure is experimentally compared against the cascade controllers in the time domain.

A. Experimental Prototype System

The manufactured prototype system of the high-precision one-DoF positioning system is shown in Fig. 6. It employs the lightweight mover held in place by an aluminum flexure structure and the VCA (VCAR0087-0062-00A, Supt Motion, CN) serving as the positioning mechanism. On top of the mover, a reflective mirror is mounted, enabling subnanometer position measurement with an interferometer (IDS3010, AttoCube Systems AG, Germany). The 3-D-printed base holding the flexure and the stator part of the VCA is mounted on a vibration-isolated laboratory table. On the right of Fig. 6, the custom-made switched current amplifier (see Fig. 1) connected to the VCA is shown. It has a maximum supply voltage of 24 V and an output current capability of up to 8 A. The controllers are implemented in a rapid prototyping system (MicroLabBox, dSPACE GmbH, Germany) with a sample time of $T_s = 1/50\,\text{kHz}$.

To identify the model parameters of the built system, the measured frequency domain transfer function from the amplifier output voltage u_{pwm} to the actuator coil current i_{act} is shown in Fig. 7. The parameters corresponding to the derived system model Fig. 2 are then determined by numerically solving a nonlinear least-squares curve fitting problem to the measured response with Matlab. The resulting model transfer function

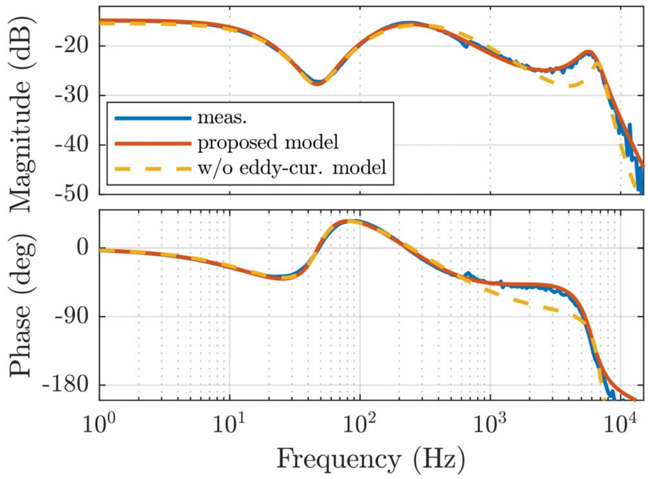

Fig. 7. Measured transfer function from u_{pwm} to i_{act} (blue). The parameters of the system model (4) are derived by solving a nonlinear least-squares curve fitting problem (red). Without the eddy-current model in Fig. 2 (constant inductance of VCA), the fitted model deviates from the measurement at higher frequencies (yellow).

TABLE II
System and Control Parameters

L_1	22.5 µH	R_p	$1{\cdot}10^8$ 1/H
R_{L1}	70 mΩ	R_m	$3.47{\cdot}10^7$ 1/H
C_1	15 µF	k_m	12.87 N/A
R_{C1}	10 mΩ	m	$47{\cdot}10^{-3}$ kg
C_{s1}	21 µF	k	$4.1{\cdot}10^3$ N/m
R_{s1}	1.35 Ω	d	8.79 N s/m
R_{vca}	5.36 Ω	T_s	1/50 kHz
G	$1.98{\cdot}10^3$ 1/Ω	$n_{\text{sens},i}$	$6{\cdot}10^{-12}$ A²/Hz
N	243	f_c	100 kHz
V_s	16 V		
Cascade-control (CC)			
$k_{I,c}$	$1.08{\cdot}10^{-4}$	T_I	$1.08{\cdot}10^{-4}$
ω_n	$33.6{\cdot}10^3$	D	0.398
ξ	0.4	k_p	$5.57{\cdot}10^3$
$k_{I,p}$	$7.78{\cdot}10^5$	k_d	8.87
T_t	$7.86{\cdot}10^{-5}$		
State-control (LQG+I)			
$\mathbf{q}$	$\left[2.6{\cdot}10^8 \;\; 4.6{\cdot}10^{15} \;\; 0.78 \;\; 0.17 \;\; 0.08 \;\; 0.003 \;\; 0.44 \;\; 2.4{\cdot}10^{19}\right]$		
R	1		
$\mathbf{q}_n$	$\left[1.62{\cdot}10^7 \;\; 1.95{\cdot}10^9 \;\; 2.5 \;\; 1.35 \;\; 1.77 \;\; 0.97 \;\; 1.75\right] \cdot 1{\cdot}10^{-6}$		
$\mathbf{r}_n$	$\left[2.3{\cdot}10^{-12} \;\; 0.57\right]$		

is represented by the red curve in Fig. 7. It is to mentioned that electrical component values which are easily accessible to measurement are fixed to their measured value to reduce the number of unknown parameters for the curve fitting problem. To evaluate the influence of the proposed eddy-current model on the modeling accuracy, in additional to the presented model, a simplified model is derived which assumes a constant VCA inductance instead of the magnetical model part in Fig. 2. The result of the numerical curve fitting result for the simplified model is shown with the yellow line in Fig. 7. Both models match the measurement very well in the lower frequency range, whereas the simple model starts deviating from the measured response at higher frequencies, at which eddy-current effects start to emerge. The identified system parameters for the full model are summarized in Table II.

To enable a fair comparison between the control structures, both the state-controller and frequency domain controllers are tuned for the same closed-loop positioning bandwidth of $f^d_{3\text{dB}} = 700$ Hz. The current loop of the CC structure is tuned to be a factor 5 faster than the outer position loop. Measurements on the experimental system reveal a phase margin of 50° for the cascade controller (data not shown).

For the tuning of the LQG+I control, a positioning uncertainty of 1nm (rms) is targeted, so $x^{\max}_{e,\text{rms}}$ is set to this value for the optimization (Section IV-D). It is to note that single-tuned parameters largely vary for different starting parameter sets used in the optimization, as the defined error function (30) is not equally sensitive to all tunable parameters. However, this variation had no visible impact on the measurement results presented in this section. Additionally, Monte Carlo simulations with the optimized parameter set are performed with variations up to 15% of the nominal plant parameters, which covers typical component variations in reality (data not shown). In all cases, the closed-loop system remains stable, giving an indication of the control systems' robustness. The optimized LQG+I parameters and the parameters of the cascade controller are given in Table II in the continuous domain and are transformed into their respective discrete counterpart for the implementation of the experimental setup.

B. Evaluation of Dynamic Error Budget

In this section, the dynamic error budget for each control structure is evaluated and compared to experimental data. The results are shown in Fig. 8, in which the measured PSD of steady-state position error and the cumulative PSD for each control scheme are plotted, together with its division into the contribution from current measurement noise ($d_{\text{sens},i}$) and external disturbances (d_{ext}). As expected from the sensitivity analysis in Fig. 5 the dominating error source for classical CC is the current measurement noise, whereas for the other cases with included KF (CC+KF, LQG+I) external disturbances are dominating. With a position uncertainty of 0.6 and 1 nm (rms), LQG+I and CC+KF show more than a tenfold improvement as compared to the 11.2 nm (rms) in the case without KF (CC). The circles in the plot represent the calculated cumulative PSD from measured-time domain data on the prototype setup, which matches the theoretical values from the dynamic error budget evaluation very well. The results show that the first control objective of a positioning uncertainty below 1 nm (rms) is clearly achieved by the optimization-based controller parameter tuning from Section IV-D.

C. System Dynamics Evaluation

In addition to the dynamic error budgeting analysis, the control structures are evaluated with respect to closed-loop system dynamics and position uncertainty in the time domain on the prototype setup. Fig. 9 shows the measured position $\tilde{x}$ and the corresponding control output u_{vca} for multiple steps on the reference input. All control schemes exhibit a similar rise time of 0.64 ms, as they are tuned for the same closed-loop bandwidth.

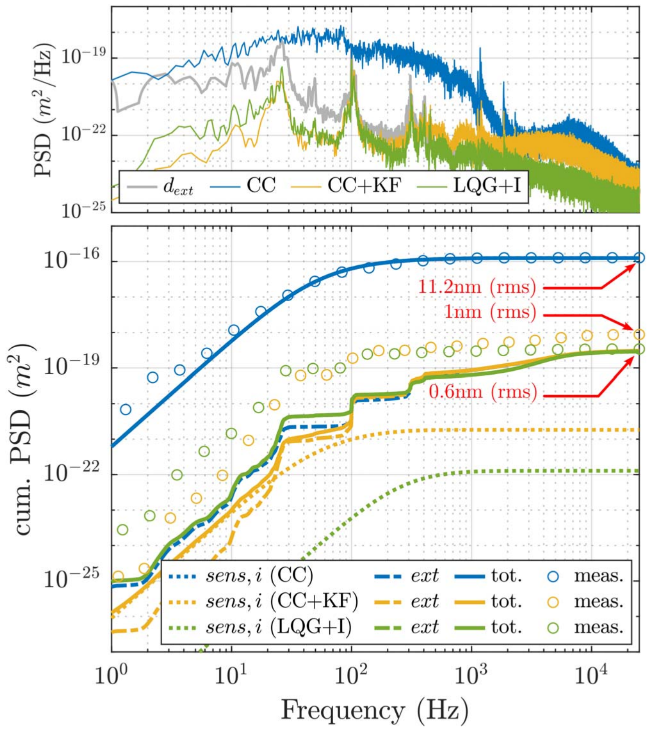

Fig. 8. Measured PSD of steady-state position error and PSD of external disturbances d_{ext} ((m^2/Hz)) (top plot). Evaluation of the dynamic error budget: theoretical and measured cumulative PSD of position error (m^2) plus division into respective error source. The corresponding rms position uncertainty for a normal distribution is marked for the end-values of the experimental data (bottom plot).

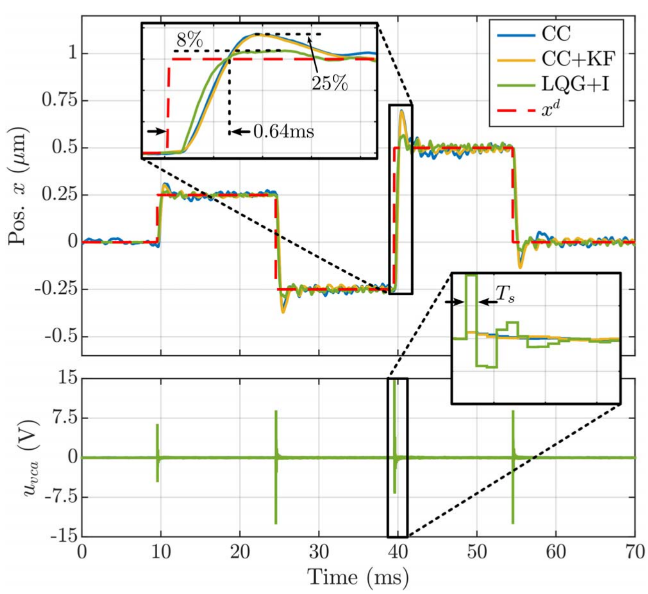

Fig. 9. Comparison of CC, CC+KF, and full-state control (LQG+I) for a steps on the reference input x^d. On the bottom, the control action u_{pwm} is plotted for all three cases.

The frequency domain controllers show an overshoot of 25%. In the dynamic comparison, the integrated state-control structure is clearly superior to the other control schemes with 8% overshoot, which corresponds to an improvement of 17%. The slight oscillations after the step result from the excitation of structural modes of the mechanical system, which emerge a few hundred hertz beyond the closed-loop bandwidth of the controllers. These modes are excited due to the step function as reference input, which could be avoided with smoother motion profiles, e.g., minimum-jerk trajectories [6].

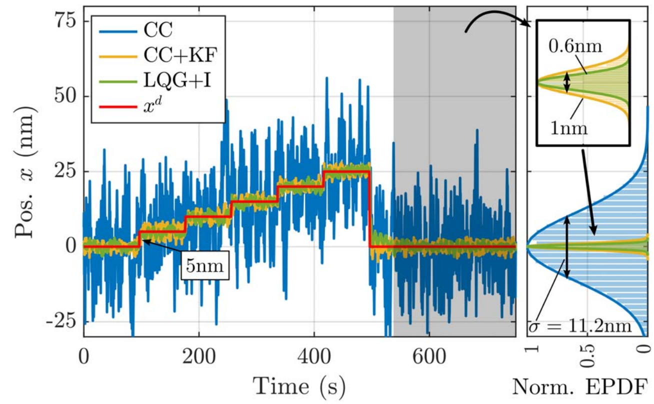

Fig. 10. Comparison of CC, CC+KF, and full-state control (LQG+I) for 5-nm steps on the reference input x^d (left). Normalized empirical probability density function (EPDF) of the steady-state position and Gaussian fit (right).

The precision and steady-state position uncertainty is evaluated in Fig. 10, in which the system response to multiple 5-nm steps at the reference input is shown. Additionally, the normalized EPDF of the position signal in the grey-shaded steady-state area is plotted together with a Gaussian fit on the right side. It confirms that the steady-state position uncertainty closely fits a normal distribution and that the values from the dynamic error budget analysis in the frequency domain match the time-domain data (refer to Fig. 8).

In summary, the effectiveness of the proposed tuning procedure for the integrated state-control (LQG+I) is successfully demonstrated on the experimental setup, achieving subnanometer positioning uncertainty and superior dynamic position tracking performance compared to the CC structures (CC, CC+KF).

VI. Conclusion

This article presents the integrated mechatronic and control design of a switched amplifier-driven high-precision positioning system. In addition to classical-cascade controllers (CC, CC+KF), a highly integrated state-control structure (LQG+I) is designed based on a comprehensive mathematical system model considering the mechanical, electrical, and magnetical systems. An optimization-based tuning process is proposed for the parameters of the LQR+I controller and KF, considering both dynamic requirements and a targeted steady-state positioning uncertainty, which is enabled by a dynamic error budget analysis in the frequency domain. The analysis reveals a steady-state positioning uncertainty of 0.6 nm (rms) for the tuned state-control structure (LQG+I), which is more than a tenfold improvement compared to a classical CC structure (11.2 nm). Experimental measurements on a built prototype system confirm these results and further demonstrate fast reference position tracking with a rise time of 0.64 ms in a step-response. While the achieved positioning uncertainties of the cascade controller with included KF (CC+KF) and LQG+I are comparable, LQG+I exhibits superior dynamic tracking performance, featuring 17% less overshoot in a step-response.

References

[1] H. Butler, "Position control in lithographic equipment [Applications of control]," *IEEE Contr. Syst.*, vol. 31, no. 5, pp. 28–47, Oct. 2011.

[2] G. Schitter, K. J. Astrom, B. E. DeMartini, P. J. Thurner, K. L. Turner, and P. K. Hansma, "Design and modeling of a high-speed AFM-scanner," *IEEE Trans. Control Syst. Technol.*, vol. 15, no. 5, pp. 906–915, Sep. 2007.

[3] L. Jonušauskas, D. Gailevičius, S. Rekštytė, T. Baldacchini, S. Juodkazis, and M. Malinauskas, "Mesoscale laser 3D printing," *Opt. Exp.*, vol. 27, no. 11, pp. 15205–15221, 2019.

[4] N. H. Vrijsen, J. W. Jansen, and E. A. Lomonova, "Comparison of linear voice coil and reluctance actuators for high-precision applications," in *Proc. 14th Int. Power Electron. Motion Control Conf. EPE-PEMC*. Piscataway, NJ, USA: IEEE Press, 2010, pp. S3-29–S3-36.

[5] R. M. Schmidt, G. Schitter, and A. Rankers, *The Design of High Performance Mechatronics: High-Tech Functionality by Multidisciplinary System Integration*. Amsterdam, The Netherlands: IOS Press, 2020.

[6] S. Ito, J. Steininger, and G. Schitter, "Low-stiffness dual stage actuator for long rage positioning with nanometer resolution," *Mechatronics*, vol. 29, pp. 46–56, Aug. 2015.

[7] D. Wertjanz, E. Csencsics, and G. Schitter, "Three-DoF vibration compensation platform for robot-based precision inline measurements on free-form surfaces," *IEEE Trans. Ind. Electron.*, vol. 69, no. 1, pp. 613–621, Jan. 2022.

[8] H. Shinno, H. Yoshioka, and H. Sawano, "A newly developed long range positioning table system with a sub-nanometer resolution," *CIRP Ann.*, vol. 60, no. 1, pp. 403–406, 2011.

[9] T. J. Teo, G. Yang, and I.-M. Chen, "A flexure-based electromagnetic nanopositioning actuator with predictable and re-configurable open-loop positioning resolution," *Precis. Eng.*, vol. 40, pp. 249–260, Apr. 2015.

[10] S. Ito and G. Schitter, "Comparison and classification of high-precision actuators based on stiffness influencing vibration isolation," *IEEE/ASME Trans. Mechatronics*, vol. 21, no. 2, pp. 1169–1178, Apr. 2016.

[11] S. Ito, S. Pirker, and G. Schitter, "Integrating PWM amplifier with the design of mechatronic systems for energy-efficient precision motion," *IEEJ J. Industry Appl.*, vol. 10, no. 2, pp. 134–141, 2021.

[12] M. Mauerer, A. Tuysuz, and J. W. Kolar, "Low-noise isolated digital shunt for precision class-d power amplifiers," *IEEE Trans. Power Electron.*, vol. 33, no. 3, pp. 1907–1910, Mar. 2018.

[13] P. Sriyotha, K. Nakamoto, M. Sugai, and K. Yamazaki, "Development of 5-axis linear motor driven super-precision machine," *CIRP Ann.*, vol. 55, no. 1, pp. 381–384, 2006.

[14] F. A. Qureshi, V. Spinu, K. Wijnands, and M. Lazar, "A real-time control system architecture for industrial power amplifiers," in *Proc. Amer. Control Conf*. Piscataway, NJ, USA: IEEE Press, 2013, pp. 4510–4515.

[15] S. J. Settels, J. van Duivenbode, and J. L. Duarte, "Impact of amplifier errors on position loop accuracy of high-precision moving stages," in *Proc. 19th Eur. Conf. Power Electron. Appl. (EPE ECCE Europe)*. Piscataway, NJ, USA: IEEE Press, 2017, pp. P.1–P.10.

[16] M. Hajiheidari, D. Xu, J. van Duivenbode, B. Vermulst, and M. Lazar, "Analysis of power amplifier contribution to the precision of motion systems," in *Proc. IEEE 17th Int. Conf. Adv. Motion Control (AMC)*. Piscataway, NJ, USA: IEEE Press, 2022, pp. 288–293.

[17] T. Riel, R. Saathof, A. Katalenic, and G. Schitter, "Noise analysis and improvement of a permanent magnet synchronous motor by dynamic error budgeting," *IFAC-PapersOnLine*, vol. 49, no. 21, pp. 339–346, 2016.

[18] A. G. Gelen and A. Atasoy, "A new method for Kalman filter tuning," in *Proc. Int. Conf. Artif. Intell. Data Process. (IDAP)*. Piscataway, NJ, USA: IEEE Press, 2018, pp. 1–6.

[19] Z. Chen, C. Heckman, S. Julier, and N. Ahmed, "Weak in the nees?: Auto-tuning Kalman filters with Bayesian optimization," in *Proc. 21st Int. Conf. Inf. Fusion (FUSION)*. Piscataway, NJ, USA: IEEE Press, 2018, pp. 1072–1079.

[20] B. M. Åkesson, J. B. Jørgensen, N. K. Poulsen, and S. B. Jørgensen, "A generalized autocovariance least-squares method for Kalman filter tuning," *J. Process Control*, vol. 18, nos. 7–8, pp. 769–779, 2008.

[21] N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics: Converters, Applications, and Design*. New York, NY, USA: John Wiley & Sons, 2003.

[22] A. Brown, J. Ross, and K. Nichols, "Time-domain simulation of mixed nonlinear magnetic and electronic systems," *IEEE Trans. Magn.*, vol. 37, no. 1, pp. 522–532, Jan. 2001.

[23] F. Baronti, A. Lazzeri, F. Lenzi, R. Roncella, R. Saletti, and S. Saponara, "Voice coil actuators: From model and simulation to automotive application," in *Proc. 35th Annu. Conf. IEEE Ind. Electron.*, pp. 1805–1810, 2009.

[24] S. Skogestad, *Multivariable Feedback Control*. New York, NY, USA: John Wiley & Sons, 2005.

[25] N. Lehtomaki, N. Sandell, and M. Athans, "Robustness results in linear-quadratic Gaussian based multivariable control designs," *IEEE Trans. Autom. Control*, vol. 26, no. 1, pp. 75–93, Feb. 1981.

[26] E. Csencsics and G. Schitter, "Parametric PID controller tuning for a fast steering mirror," in *Proc. IEEE Conf. Control Technol. Appl. (CCTA)*. Piscataway, NJ, USA: IEEE Press, 2017, pp. 1673–1678.

[27] L. Jabben and J. Van Eijk, "Performance analysis and design of mechatronic systems," *Mikroniek*, vol. 51, no. 2, pp. 5–12, 2011.

[28] A. Okyay, K. Erkorkmaz, and M. B. Khamesee, "Mechatronic design, actuator optimization, and control of a long stroke linear nano-positioner," *Precis. Eng.*, vol. 52, pp. 308–322, Apr. 2018.

[29] T. Riel, R. Saathof, A. Katalenic, S. Ito, and G. Schitter, "Noise analysis and efficiency improvement of a pulse-width modulated permanent magnet synchronous motor by dynamic error budgeting," *Mechatronics*, vol. 50, pp. 225–233, Apr. 2018.

[30] W. A. Gardner, *Introduction to Random Processes*. New York, NY, USA: McGraw-Hill, 1989.

[31] Texas Instruments, "Noise analysis in operational amplifier circuits," 2007. Accessed: Nov. 20, 2023. [Online]. Available: https://www.ti.com/lit/an/slva043b/slva043b.pdf?ts=1724097135837

Daniel Pechgraber received the M.Sc. degree in electrical engineering from TU Wien, Vienna, Austria, in 2022.

He is a Doctoral Researcher with the Advanced Mechatronic Systems Group, Automation and Control Institute (ACIN), TU Wien. His research interests include high-performance mechatronic systems design, high-precision motion control of linear and nonlinear systems, as well as power electronics design for precision motion systems.

Johannes Wiesböck received the M.Sc. degree in electrical engineering from TU Wien, Vienna, Austria, in 2021.

He is a Doctoral Researcher with the Advanced Mechatronic Systems Group, Automation and Control Institute (ACIN), TU Wien. His research interests include the design and system integration of high-precision optomechatronic measurement systems and optical sensing technologies for material analysis.

Ernst Csencsics (Member, IEEE) received the M.Sc. and Ph.D. (sub auspiciis) degrees in electrical engineering from TU Wien, Vienna, Austria, in 2014 and 2017, respectively.

He is an Assistant Professor in metrology systems with the Advanced Mechatronic Systems Group, Automation and Control Institute (ACIN), TU Wien. His research interests include high-performance mechatronic systems, the development of holistic methods for multidisciplinary system design and integration, optomechatronic measurement and imaging systems, precision engineering, and robot-based in-line measurement systems.

Dr. Csencsics received the journal best paper award of IEEE/ASME TRANSACTIONS ON MECHATRONICS (2018) and the best student paper award at the American Control Conference (2016).

Georg Schitter received the M.Sc. degree in electrical engineering from TU Graz, Graz, Austria, in 2000, and the M.Sc. degree in information technology and Ph.D. degree in technical sciences from ETH Zurich, Zurich, Switzerland, in 2004.

He is a Professor in advanced mechatronic systems with the Automation and Control Institute (ACIN), TU Wien, Vienna, Austria. His research interests include high-performance mechatronic systems, particularly for applications in the high-tech industry, scientific instrumentation, and mechatronic imaging systems, such as Atomic Force Microscopy (AFM), scanning laser and Light Detection and Ranging (LIDAR) systems, telescope systems, adaptive optics, and lithography systems for semiconductor industry.

Dr. Schitter received the journal best paper award of IEEE/ASME TRANSACTIONS ON MECHATRONICS (2018), *International Federation of Automatic Control (IFAC) Mechatronics* (2008–2010), *Asian Journal of Control* (2004–2005), and the 2013 IFAC Mechatronics Young Researcher Award. He served as an Associate Editor for *IFAC Mechatronics*, *Control Engineering Practice*, and IEEE TRANSACTIONS ON MECHATRONICS.